Stock Price Prediction using SVM

Regression model to predict next-day stock closing price using historical OHLCV data, built with scikit-learn.

Overview

This project uses 5 years of real daily stock data (Apple Inc. - AAPL) to predict the next trading day's closing price using Support Vector Regression (SVR).

Approach

- **Data source**: real historical OHLCV (Open, High, Low, Close, Volume) data pulled via Yahoo Finance
- **Feature engineering**: added 5-day and 10-day moving averages and rolling volatility as features, since raw price data alone is a weak predictor
- **Train/test strategy**: chronological split (not random shuffle) — trained on earlier dates, tested on most recent dates, to properly simulate real-world forecasting and avoid lookahead bias
- **Model**: Support Vector Regression (SVR) with RBF kernel
- **Hyperparameter tuning**: GridSearchCV with 5-fold cross-validation across C, gamma, and kernel parameters to reduce overfitting
- **Evaluation metrics**: R^2 score, RMSE, MAE, and MAPE for a complete view of prediction error

Results

(Run the notebook to generate your own numbers, then update this section)

- R^2 Score: 0.XX
- RMSE: $X.XX
- MAE: $X.XX
- MAPE: X.X%

Tech Stack

Python, yfinance, Pandas, NumPy, Scikit-learn (SVR, GridSearchCV), Matplotlib

How to Run

1. Open `stock_price_prediction.py` in Google Colab or Jupyter
2. Run all cells — data downloads automatically via yfinance
3. Review printed metrics and the actual-vs-predicted price chart

Future Improvements

- Compare against LSTM/time-series deep learning models
- Add more technical indicators (RSI, MACD, Bollinger Bands)
- Test across multiple stocks/sectors for generalization